



Enterprise COBOL QA Automation

Transform Your **COBOL Testing** with AI Precision

WMA-TestGen reads your Business Systems Design and COBOL modules, then generates a complete, mathematically verified test suite — in seconds, not weeks.

▶ **See Live Demo**

ⓘ **How It Works**



90%

Reduction in Test Design Time

100%

Business Rule Coverage Guaranteed

20+

Test Cases per Module

\$162k saved

Avg. Savings per Enhancement

THE CHALLENGE

COBOL Testing Is Broken

Legacy annuity systems process billions in premium each year — yet QA relies on manual, error-prone, undocumented test design.



Weeks of Manual Work

A single enhancement BSD takes 1–2 weeks to manually translate into a test suite. With dozens of modules releasing quarterly, teams are perpetually behind.



Missed Edge Cases

Boundary values, COMP-3 rounding, free-look periods, qualified-plan tax rules — manual test designers miss critical edge cases that slip into production.





Auditors demand a clear link from every business rule to a test case and expected result. Manual spreadsheets can't provide it — compliance risk is real.

THE SOLUTION

AI That Understands COBOL Logic

WMA-TestGen reads your actual code — not a description of it — and produces verified, traceable test cases with computed expected values.



Reads BSD + COBOL Source

Ingests your Business Systems Design and actual .cbl / .cpy files. No manual input, no proxy documents.



Computes Expected Values from Code

Traces every COMPUTE statement. Shows arithmetic step-by-step. COMP-3 precision, ROUNDED clauses — all honored.



Full Traceability Matrix

Every test case cites the BR it covers. Every BR points to test IDs. Every error code has a negative test. Ready for audit.



Formal Documentation Auto-Generated

Outputs a complete test plan with RTM, environment setup, execution notes, and sign-off block — attach directly to your change record.





```
// WMA-TestGen Output - TC-010
Test ID : TC-010
```

Title : Q-plan above free allowance

Traces : BR-1,3,4,5,7,8

```
// Expected Calculation
```

Free allow = 100000 × 0.10 = 10000.00

Excess = 20000 - 10000 = 10000.00

Surr chg = 10000 × 0.05 = 500.00

Tax w/h = 19500 × 0.10 = 1950.00

Net WD = 20000-500-1950= 17550.00

Return Code = '0000' ✓ PASS

PROCESS

Six Phases. One Complete Suite.

A deterministic, standards-driven process that runs end-to-end — from reading your BSD to emitting signed-off documentation.

1

Ingest & Inventory

Reads every module, copybook, and the BSD. Maps COPY relationships, CALL chains, and linkage sections. Flags any missing inputs explicitly.

Input Analysis

2

Data Domain Extraction

Extracts PIC clauses, COMP-3 fields, 88-levels, signed/unsigned fields, and implied decimals into a typed field catalog.

Field Catalog



Rule & Path Analysis

Extracts every BR from the BSD and every branch (IF / EVALUATE / 88-level) from the code. Reconciles them — flags any gaps.

Gap Detection

Test Case Design

Applies EP, BVA, Decision-table, State-transition, and Annuity Domain techniques. Guarantees every BR and every error code is covered.

Coverage Design

5

Expected-Result Computation

Executes COBOL logic symbolically. Shows arithmetic step-by-step. Respects ROUNDED, truncation, and COMP-3 precision. Never guesses.

Math Verification

6

Self-Review Pass

Critiques its own suite: checks BR ↔ test coverage table for blank cells, validates input values fit PIC clauses, adds missing cases.

Quality Assurance

INTERACTIVE DEMO

See It in Action

This is the actual BSD for a real annuity withdrawal enhancement (WMA-ANN-2026-0142). Click Generate to see WMA-TestGen produce the test suite.

📄 BSD Input — WMA-ANN-2026-0142



- Free-Withdrawal Allowance
& Qualified-Plan Tax Withholding

Module: WITHDRAW.cbl

BUSINESS RULES

BR-1: Free allowance = 10% of
account value per policy year.

BR-2: Prior YTD withdrawals reduce
the remaining free allowance.

✓ Generated — 20 Test Cases

Generated Test Suite

STATUS='I' · PLAN='N' · AV=50000.00 · WD=50000.01

50000.01 > 50000.00 → reject. No arithmetic performed.

TC-017

0000

NQ — Zero surrender charge rate; no fee on excess

Technique: EP (rate=0)

STATUS='I' · PLAN='N' · AV=100000.00 · WD=50000.00 · PCT=0%

Free=10000 | Excess=40000 | Surr=40000×0=0 | Tax=0 | Net=50000 | NewAV=50000

✓ 20 test cases generated · 8/8 BRs covered · 3/3 error codes · 5 techniques applied



PILOT RESULTS

Real Output. Verified Math.

Applied to BSD WMA-ANN-2026-0142 (partial withdrawal enhancement).
Generated in one pass.



Coverage Achieved



Sample Test Cases

ID	Title	RC
TC-001	Policy not INFORCE (Pending)	E301
TC-005	NQ — within free allowance	0000
TC-010	Q — above allowance, surr+tax	0000
TC-015	WD = account value exactly	0000
TC-016	WD = AV + \$0.01 (E303)	E303



Verified Calculation — TC-010: Qualified Plan, Above Free Allowance



Excess = 20,000.00 - 10,000.00 = 10,000.00

Surr Charge = 10,000.00 × 0.05 = 500.00

Tax Withheld = (20,000.00 - 500.00) × 0.10 = 1,950.00

Net Withdrawal = 20,000.00 - 500.00 - 1,950.00 = 17,550.00

New Acct Value = 100,000.00 - 20,000.00 = 80,000.00

✓ LK-SURR-CHARGE-OUT=500.00 · LK-TAX-WITHHELD=1950.00 · LK-NET-WD=17550.00 · LK-RETURN-CODE='0000'

RETURN ON INVESTMENT

The Numbers Speak

Measured against a team of 3 COBOL QA engineers handling 8 module enhancements per quarter.

BEFORE

- ✗ 2 weeks per module
- ✗ ~60% BR coverage
- ✗ No traceability
- ✗ Manual arithmetic
- ✗ Audit gaps
- ✗ \$22,400/module

AFTER

- ✓ 2 hours per module
- ✓ 100% BR coverage
- ✓ Full RTM
- ✓ Verified math
- ✓ Audit-ready docs
- ✓ ~\$2,200/module

90%

Reduction in time-to-test, freeing engineers for higher-value work

0

Business rules without a covering test case — guaranteed by design

ROLLOUT PLAN

Production-Ready in 4 Weeks

A low-risk, phased rollout that proves value in week one before any broader commitment.

WEEK 1 — PILOT

One Module, Full Run

Select one active enhancement BSD. Run WMA-TestGen end-to-end. Compare output to your team's manual test suite. Measure delta.

WEEK 2 — CALIBRATE

Tune to Your Conventions

Align message text literals, return-code formats, and field naming conventions to match your existing COBOL standards and change-record templates.

WEEK 3 — INTEGRATION

Workflow Integration

Integrate into the BSD intake process. Developer submits BSD + modules; WMA-TestGen returns the suite. Team reviews, approves, executes.

WEEK 4 — PRODUCTION

Full Production Deployment



NEXT STEP

Ready to Eliminate Manual Test Design Forever?

Schedule a 30-minute live walkthrough. We'll run WMA-TestGen on one of your actual BSDs — live, on the call.

 Schedule a Demo

 View on GitHub